Chapter 8 Review Questions

1. A firewall sits between the organization's network and the public network, controlling packet access to and from the network. An intrusion detection system performs "deep packet inspection," alerting the network administrators about suspicious activity.

2. User's laptop and a web server; (ii) two routers; (iii) two DNS name servers.

3. It comes from the secrecy of the key only.

4. When an intruder knows some of the (plaintext, ciphertext) pairings, we refer to this as a known-plaintext attack on the encryption scheme. In a chosen-plaintext attack, the intruder is able to choose the plaintext message and obtain its corresponding ciphertext form.

5. An 16-block cipher has 2^{16} possible input blocks. Each mapping is a permutation of the 2^{16} input blocks; so there are $2^{16}!$ possible mappings; so there are $2^{16}!$ possible keys.

6. If each user wants to communicate with N other users, then each pair of users must have a shared symmetric key. There are N*(N-1)/2 such pairs and thus there are N*(N-1)/2 keys. With a public key system, each user has a public key which is known to all, and a private key (which is secret and only known by the user). There are thus 2N keys in the public key system.

7. a mod n = 17 , b mod n = 6. So $(a \times b) \bmod n = 17 \times 6 = 102$

8. 201

9. One requirement of a message digest is that given a message M, it is very difficult to find another message M' that has the same message digest and, as a corollary, that given a message digest value it is difficult to find a message M'' that has that given message digest value. We have "message integrity" in the sense that we have reasonable confidence that given a message M and its signed message digest that the message was not altered since the message digest was computed and signed. This is not true of the Internet checksum, where we saw in Figure 7.18 that it easy to find two messages with the same Internet checksum.

10. No. This is because a hash function is a one-way function. That is, given any hash value, the original message cannot be recovered (given h such that h=H(m), one cannot recover m from h).

11. This is scheme is clearly flawed. Trudy, an attacker, can first sniff the communication and obtain the shared secret s by extracting the last portion of digits from H(m)+s.

Trudy can then masquerade as the sender by creating her own message t and send (t, H(t)+s).

12. Suppose Bob sends an encrypted document to Alice. To be verifiable, Alice must be able to convince herself that Bob sent the encrypted document. To be non-forgeable, Alice must be able to convince herself that only Bob could have sent the encrypted document (e.g.,, no one else could have guessed a key and encrypted/sent the document) To be non-reputable, Alice must be able to convince someone else that only Bob could have sent the document. To illustrate the latter distinction, suppose Bob and Alice share a secret key, and they are the only ones in the world who know the key. If Alice receives a document that was encrypted with the key, and knows that she did not encrypt the document herself, then the document is known to be verifiable and non-forgeable (assuming a suitably strong encryption system was used). However, Alice cann*ot* convince someone else that Bob must have sent the document, since in fact Alice knew the key herself and could have encrypted/sent the document.

13. A network administrator could actually accomplish this by physically visiting each of the routers. Or, if each router has its own public key, the network administrator could distribute the authentication key to any one of the routers by encrypting it with the router's public key and then sending the encrypted key over the network to the router.

14. IPsec and SSL.

15. For a MAC-based scheme, Alice would have to establish a shared key with each potential recipient. With digital signatures, she uses the same digital signature for each recipient; the digital signature is created by signing the hash of the message with her private key. Digital signatures are clearly a better choice here.

16. The purpose of the nonce is to defend against the replay attack.

17. Once in a lifetimes means that the entity sending the nonce will never again use that value to check whether another entity is "live".

18. In a man-in-the-middle attack, the attacker puts himself between Alice and Bob, altering the data sent between them. If Bob and Alice share a secret authentication key, then any alterations will be detected.

19. PGP. It uses digital signatures for authentication and integrity.

20. False. SSL uses implicit sequence numbers.

21. The purpose of the random nonces in the handshake is to defend against the connection replay attack.

22. True. The IV is always sent in the clear. In SSL, it is sent during the SSL handshake.

23. After the client will generate a pre-master secret (PMS), it will encrypt it with Alice's public key, and then send the encrypted PMS to Trudy. Trudy will not be able to decrypt the PMS, since she does not have Alice's private key. Thus Trudy will not be able to determine the shared authentication key. She may instead guess one by choosing a random key. During the last step of the handshake, she sends to Bob a MAC of all the handshake messages, using the guessed authentication key. When Bob receives the MAC, the MAC test will fail, and Bob will end the TCP connection.

24. False. Typically an IPsec SA is first established between Host A and Host B. Then all packets in the stream use the SA.

25. False. IPsec will increment the sequence number for every packet it sends.

26. SSL does not mandate a specific symmetric key algorithm, a specific public-key algorithm, or a specific MAC. Instead, SSL allows to agree on the cryptographic algorithms at the beginning of the SSL session, during the handshake phase.

27. 11100111

28. No. In WEP, an IV is sent in the clear in every frame.

29. Filter table and connection table. The connection table keeps track of connections, allowing for a finer degree of packet filtering.

30. True

31. True

32. If there isn't a packet filter, than users inside the institution's network will still be able to make direct connections to hosts outside the institution's network. The filter forces the users to first connect to the application gateway.

33. True

Chapter 8 Problems

Problem 1

Encoding of "This is a secret message" Would be "uasi si m ishlgc hciimzc." Decoding of the message "fsgg ash" is "kill him."

Problem 2

If Trudy knew that the words "bob" and "alice" appeared in the text, then she would know the ciphertext for b,o,a,l,i,c,e (since "bob" is the only palindrome in the message, and "alice" is the only 5-letter word. If Trudy knows the ciphertext for 7 of the letters, then she only needs to try 19!, rather than 26!, plaintext-ciphertext pairs. The difference between 19! and 26! is 26*25*24...*20, which is 3315312000, or approximately 10^9.

Problem 3

Every letter in the alphabet appears in the phrase "The quick fox jumps over the lazy brown dog." Given this phrase in a chosen plaintext attack (where the attacker has both the plain text, and the ciphertext), the Caesar cipher would be broken - the intruder would know the ciphertext character for every plaintext character. However, the Vigenere cipher does not alway translate a given plaintext character to the same ciphertext character each time, and hence a Vigenere cipher would not be immediately broken by this chosen plaintext attack.

Problem 4

a) The output is equal to 00000101 repeated eight times.
b) The output is equal to 00000101 repeated seven times + 10000101.
c) We have (ARBRCR)R = CBA, where A, B, C are strings, and R means inverse operation. Thus:
    1. For (a), the output is 10100000 repeated eight times;
    2. For (b), the output is 10100001 + 10100000 repeated seven times.

Problem 5

a) There are 16 tables. Each table has 2^{16} entries. Each entry has 8 bits.
number of tables × size of each table × size of each entry = $16 \times 2^{16} \times 8 = 2^{23}$ bits

b) There are 2^{128} entries. Each entry has 128 bits. Thus, size = $2^{128} \times 128 = 2^{135}$.

Problem 6

a) 100100100 ==> 011011011
b) Trudy will know the three block plaintexts are the same.
c) c(i) = KS(m(i) XOR c(i-1))
   c(1) = KS(100 XOR 111) = KS (011) = 100
   c(2) = KS(100 XOR 100) = KS (000) = 110
   c(1) = KS(100 XOR 110) = KS (010) = 101

Problem 7

a) We are given $p = 5$ and $q = 7$. We thus have $n = 35$. Choose $e = 29$ since 29 and $(p-1)\times(q-1) = 24$ have no common factors. Choose $d = 5$ also so that $e \times d = 145$ and thus $e \times d - 1 = 144$ is exactly divisible by 24. We can now perform the RSA encryption and decryption using $n = 35$, $e = 29$ and $d = 5$.

m	$c = m^{29} \bmod 35$	$m = c^5 \bmod 35$
12	17	12
19	24	19
27	27	27

b) Take $p = 41$, $q = 53$, $n = p \times q = 2173$, $z = (p-1)(q-1) = 2080$, $e = 57$, $d = 73$
   Message $m = 1834$
   Ciphertext $c = 1834^{57} \bmod 2173 = 874$

Problem 8

$p = 7$, $q = 13$
(a) $n = p \times q = 91$, $z = (p-1)(q-1) = 72$
(b) $e = 17$ is less than n and has no common factors with z.
(c) $d = 17$
(d) $m = 9$, $m^e \bmod n = 81$

Problem 9

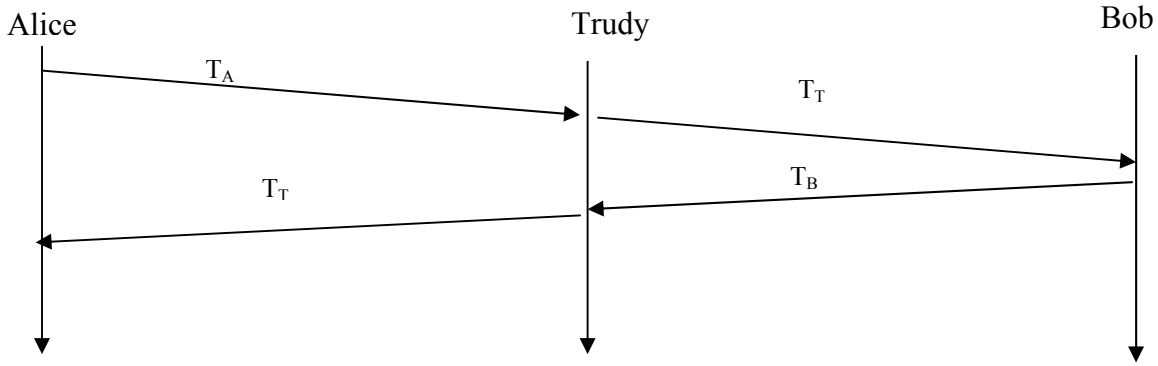
	Alice	Bob
secrect key:	S_A	S_B
public key:	$T_A = (g \text{^} S_A) \bmod p$	$T_B = (g \text{^} S_B) \bmod p$
shared key:	$S = (T_B \text{^} S_A) \bmod p$	$S' = (T_A \text{^} S_B) \bmod p$

a) S = (TB^SA) mod p = ((g^SB mod p)^SA) mod p = (g^(SBSA)) mod p
   = ((g^SA mod p)^SB) mod p = (TA^SB) mod p = S'

(b and c) p = 11, g = 2

	Alice	Bob
secrect key:	$S_A = 5$	$S_B = 12$
public key:	$T_A = (g \hat{} S_A) \bmod p = 10$	$T_B = (g \hat{} S_B) \bmod p = 4$
shared key:	$S = (T_B \hat{} S_A) \bmod p = 1$	$S' = (T_A \hat{} S_B) \bmod p = 1$

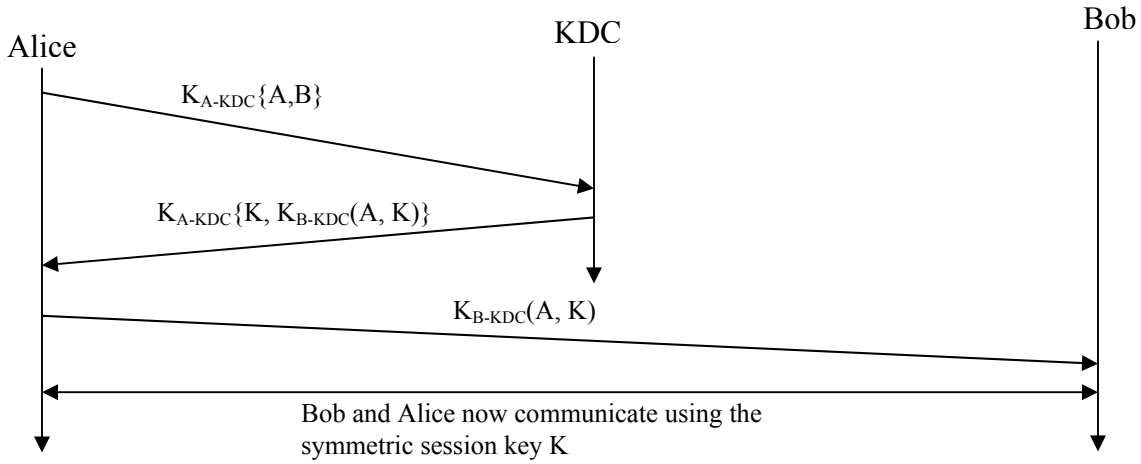
d)



The Diffie-Hellman public key encryption algorithm is possible to be attacked by man-in-the-middle.

1. In this attack, Trudy receives Alice's public value (T_A) and sends her own public value (T_T) to Bob.
2. When Bob transmits his public value (T_B), Trudy sends her public key to Alice (T_T).
3. Trudy and Alice thus agree on one shared key (S_{AT}) and Trudy and Bob agree on another shared key (S_{BT}).
4. After this exchange, Trudy simply decrypts any messages sent out by Alice or Bob by the public keys S_{AT} and S_{BT}.

Problem 10



Problem 11

The message

I	O	U	1
9	0	.	9
0	B	O	B

has the same checksum

Problem 12

The remedy is Cipher Block Chaining (CBC). The basic idea is to send only on random value along with the very first message, and then have the sender and receiver use the computed coded blocks in place of the subsequent random number.

Problem 13

The file is broken into blocks of equal size. For each block, calculate the hash (for example with MD5 or SHA-1). The hashes for all of the blocks are saved in the .torrent file. Whenever a peer downloads a block, it calculates the hash of this block and compares it to the hash in the .torrent file. If the two hashes are equal, the block is valid. Otherwise, the block is bogus, and should be discarded.

Problem 14

No.

Problem 15

Bob does not know if he is talking to Trudy or Alice initially. Bob and Alice share a secret key K_{A-B} that is unknown to Trudy. Trudy wants Bob to authenticate her (Trudy) as Alice. Trudy is going to have Bob authenticate himself, and waits for Bob to start:

1. **Bob-to-Trudy: "I am Bob"** Commentary: Bob starts to authenticate himself. Bob's authentication of himself to the other side then stops for a few steps.
2. **Trudy-to-Bob: "I am Alice"** Commentary: Trudy starts to authenticate herself as Alice
3. **Bob-to-Trudy: "R"** Commentary: Bob responds to step 2 by sending a nonce in reply. Trudy does not yet know $K_{A-B}(R)$ so she can not yet reply.
4. **Trudy-to-Bob: "R"** Commentary: Trudy responds to step 1 now continuing Bob's authentication, picking as the nonce for Bob to encrypt, *the exact same value that Bob sent her to encrypt in Step 3.*
5. **Bob-to-Trudy: "**$K_{A-B}(R)$" Bob completes his own authentication of himself to the other side by encrypting the nonce he was sent in step 4. Trudy now has $K_{A-B}(R)$. (Note: she does not have, nor need, K_{A-B}

6. **Trudy-to-Bob: "**$K_{A-B}(R)$" Trudy completes her authentication, responding to the R that Bob sent in step 3 above with $K_{A-B}(R)$. Since Trudy has returned the properly encrypted nonce that Bob send in step 3, Bob thinks Trudy is Alice!

Problem 16

This wouldn't really solve the problem. Just as Bob thinks (incorrectly) that he is authenticating Alice in the first half of Figure 7.14, so too can Trudy fool Alice into thinking (incorrectly) that she is authenticating Bob. The root of the problem that neither Bob nor Alice can tell is the public key they are getting is indeed the public key of Alice of Bob.

Problem 17

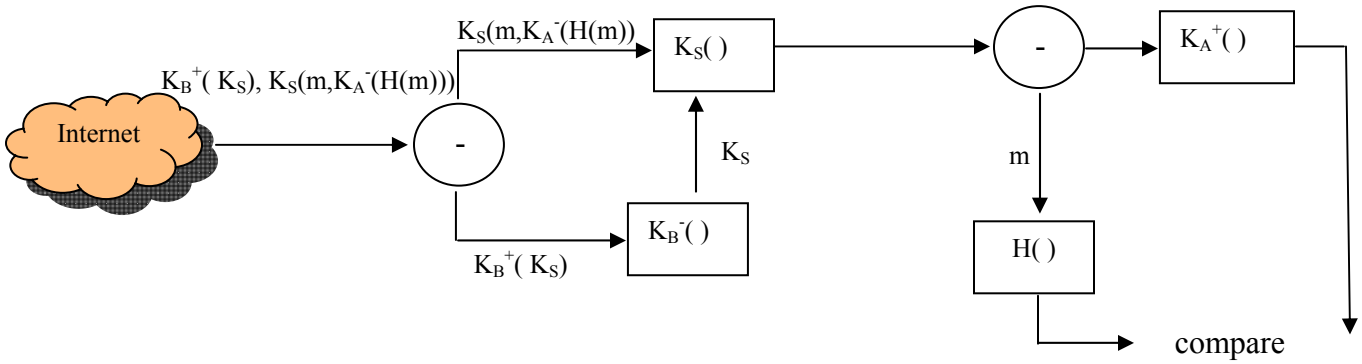


Figure: Operations performed by Bob for confidentiality, integrity, and authentication

Problem 18

a) No, without a public-private key pair or a pre-shared secret, Bob cannot verify that Alice created the message.
b) Yes, Alice simply encrypts the message with Bob's public key and sends the encrypted message to Bob.

Problem 19

a) Client
b) IP: 216.75.194.220, port: 443
c) 283
d) 3 SSL records
e) Yes, it contains an encrypted master secret
f) First byte: bc; Last byte: 29
g) 6

Problem 20

Again we suppose that SSL does not provide sequence numbers. Suppose that Trudy, a woman-in-the-middle, deletes a TCP segment. So that Bob doesn't anything, Trudy needs to also adjust the sequence numbers in the subsequent packets sent from Alice to Bob, and the acknowledgment numbers sent from Bob to Alice. The result will be that Bob will, unknowingly, be missing a packet's worth of bytes in the byte stream.

Problem 21

Message integrity can solve the issue. When router B receives a link-state message from router A, router B should verify that router A actually created the message and, further, that no one tampered with the message in transit.

Problem 22

a) F
b) T
c) T
d) F

Problem 23

For Alice to verify the signature, she must also have a copy of the key, in which case the key would not be unique to Bob.

Problem 24

a) Since IV = 11, the key stream is 111110100000 …….

   Given, m = 10100000
   Hence, ICV = 1010 XOR 0000 = 1010
   The three fields will be:

   IV: 11
   Encrypted message: 10100000 XOR 11111010 = 01011010
   Encrypted ICV: 1010 XOR 0000 = 1010

b) The receiver extracts the IV (11) and generates the key stream 111110100000 …….

   XORs the encrypted message with the key stream to recover the original message:
   01011010 XOR 11111010 = 10100000

XORs the encrypted ICV with the keystream to recover the original ICV:

1010 XOR 0000 = 1010

The receiver then XORs the first 4 bits of recovered message with its last 4 bits:
1010 XOR 0000 = 1010 (which equals the recovered ICV)

c) Since the ICV is calculated as the XOR of first 4 bits of message with last 4 bits of message, either the 1st bit or the 5th bit of the message has to be flipped for the received packet to pass the ICV check.

d) For part (a), the encrypted message was 01011010
Flipping the 1st bit gives, 11011010
Trudy XORs this message with the keystream:
11011010 XOR 11111010 = 00100000

If Trudy flipped the first bit of the encrypted ICV, the ICV value received by the receiver is 0010
The receiver XORs this value with the keystream to get the ICV:
0010 XOR 0000 = 0010
The receiver now calculates the ICV from the recovered message:
0010 XOR 0000 = 0010 (which equals the recovered ICV and so the received packet passes the ICV check)

Problem 25

Filter Table:

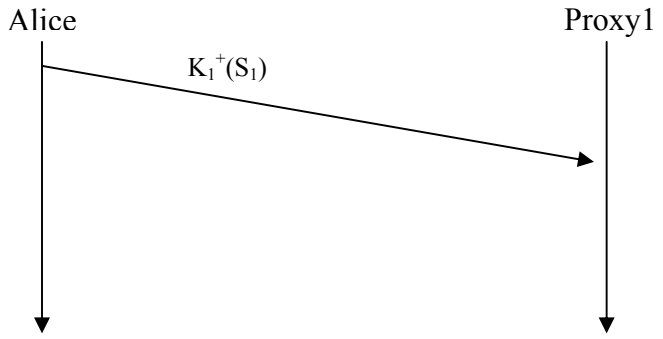
Action	Source Address	Dest address	Protocol	Source port	Dest port	Flag bit	Check connection
allow	222.22/16	outside of 222.22/16	TCP	> 1023	23	any	
allow	outside of 222.22/16	222.22/16	TCP	23	> 1023	ACK	x
Allow	outside of 222.22/16	222.22.0.12	TCP	>1023	80	Any	
Allow	222.22.0.12	outside of 222.22/16	TCP	80	>1023	Any	
deny	All	all	all	all	all	All	

Connection Table:

Source address	Dest address	Source port	Dest port
222.22.1.7	37.96.87.123	12699	23
222.22.93.2	199.1.205.23	37654	23
222.22.65.143	203.77.240.43	48712	23

Problem 26

a)



b)

